Predicting Psychiatric Illness Risk from Brain Structure via Genetic Associations

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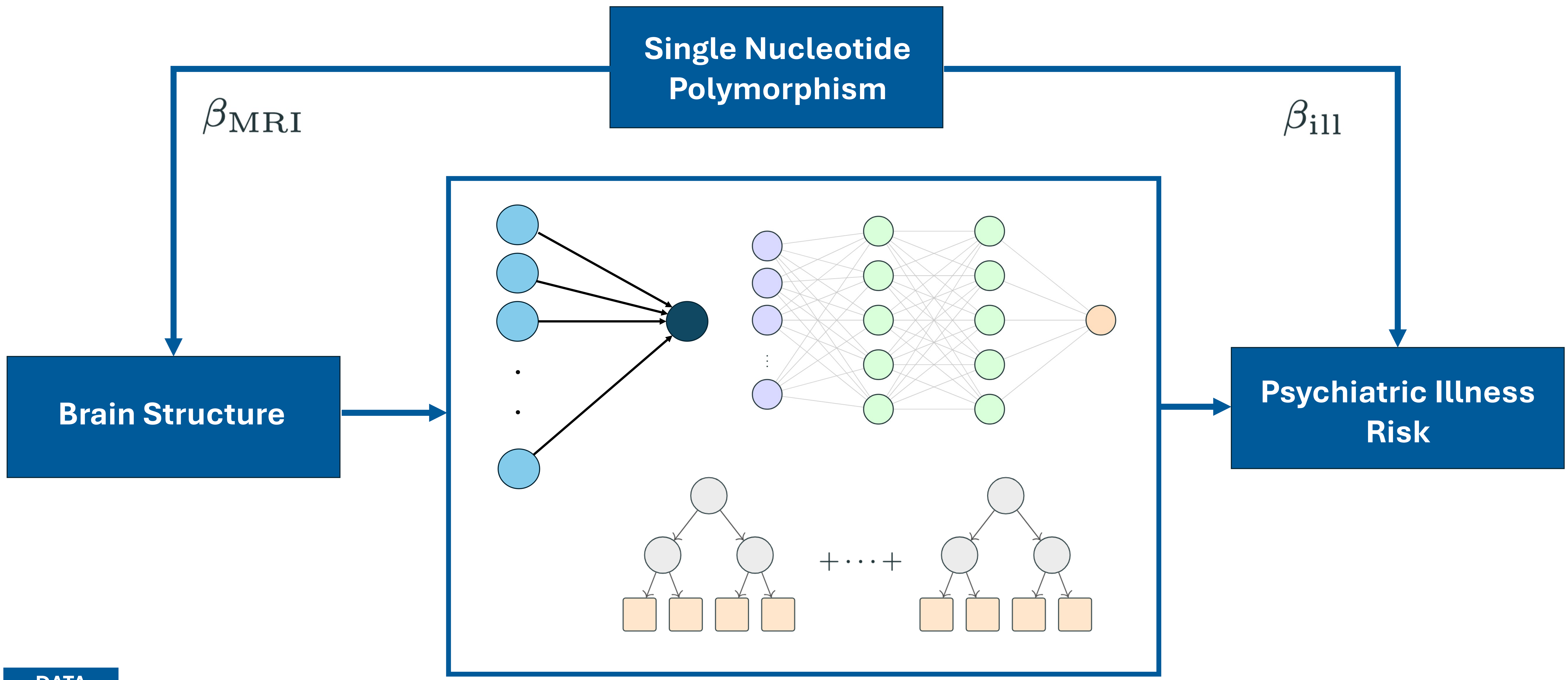

DATA

Sources of MRI-GWAS and Ilness-GWAS

Phenotype	Consortium	N_{case}	N_{control}
MRI-derived phenotypes (≈1,000 brain features)			
Structural MRI (UK Biobank)	UKB	33,000	-
Psychiatric illnesses			
SCZ	PGC-SWG	53,386	77,258
BIP	PGC	59,287	781,022
MDD	PGC-MDD	59,851	113,154
ADHD	PGC-iPSYCH	38,691	186,843
AZ	IGAP-UKB	71,88	383,378
OCD	PGC-OCD	23,493	1,114,613

Size of dataset filtered at different p_{ill} thresholds.

Illness	$N_{<10^{-4}}$	$N_{<10^{-3}}$	$N_{<10^{-2}}$	$N_{<10^{-1}}$
SCZ	1,941	5,249	17,731	67,702
BIP	1,071	3,245	11,608	45,559
ADHD	578	2,198	9,062	36,951
MDD	369	1,738	9,283	43,899
AZ	338	1,594	9,784	54,271
OCD	255	1,275	6,704	32,664

Rows = SNPs | Features = MRI z-scores | Target = illness z-score.

$$\left[\begin{array}{cccc|c} z_{1,1} & z_{1,2} & \cdots & z_{1,m} & z_{\mathrm{ill},1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,m} & z_{\mathrm{ill},2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{N,1} & z_{N,2} & \cdots & z_{N,m} & z_{\mathrm{ill},N} \end{array}\right]$$

EXPERIMENTS

	No Filter on Brain Structure significance	Filter on Brain Structure significance	Questions
No Filter on psychiatric risk significance	EXPERIMENT 1 Baseline with all data Worst performance	EXPERIMENT 3 Increase of signal in brain structure	Does more signal in brain structure lead to better performance?
Filter on psychiatric risk significance	EXPERIMENT 2 Increase of signal in psychiatric risk	EXPERIMENT 4 Most signal available	Does more signal in psychiatric risk lead to better performance?